

A new solar radiation database for estimating PV performance in Europe and Africa

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Abstract

The Photovoltaic Geographical Information System (PVGIS) is a web application for the estimation of the performance of photovoltaic (PV) systems in Europe and Africa, which has become widely used by the PV community in Europe. We here present the results of adapting the solar radiation data calculated from satellite data in the Climate Monitoring Satellite Application Facility (CM-SAF) to PVGIS. The CM-SAF solar radiation database is characterized by very low overall bias and shows good accuracy at validation sites. The application to PVGIS brings important improvements relative to the existing solar radiation databases within PVGIS.

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1. Introduction

Knowledge of the solar radiation arriving at the surface of the Earth is important for many different fields of study. Meteorological and climatological studies rely on knowledge of the solar radiation field over large areas in both the short and long term. Planning of solar energy systems requires data on solar irradiation at the site where the system will be installed. Local knowledge of the solar radiation intensity is also used in fields as varied as biological research and architecture.

Satellite data have been used to estimate the solar irradiance by many research groups and companies, using different approaches and employing a number of different satellite-based data sets. For instance, the Heliosat method or respective derivatives are well established and applied within the solar energy community. One of the first

descriptions of the Heliosat method was [Cano et al. \(1986\)](#). Heliosat uses the satellite signal of cloud reflection in the visible channel to retrieve the cloud transmission and subsequently the solar surface irradiance by application of a clear sky model. The physical basics of the Heliosat method, especially the origins in the radiation balance of the Earth-atmosphere system are discussed in [Hay \(2003\)](#).

Surface solar irradiance derived with the Heliosat method have been extensively validated against the ground-based solar radiation measurements ([Perez et al., 1997, 2001](#); [Rigollier et al., 2004](#); [Ineichen et al., 2009](#)). The relative standard deviation between the hourly mean surface solar irradiances derived using the Heliosat algorithms and the ground-based measurements is typically 20–25% (see for instance [Ineichen and Perez, 1999](#); [Zelenka et al., 1999](#); [Dagestad, 2004](#)). The accuracy, measured by the mean absolute deviation of monthly means, is usually better than 10 W/m² ([Posselt et al., 2011](#)).

An overview about solar energy applications of satellite based solar irradiance data is given in [Hammer et al.](#)

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Nomenclature

Abbreviations

BSRN	Baseline Surface Radiation Network
CM-SAF	Climate Monitoring Satellite Application Facility
DNI	direct normal irradiance
ESRA	European Solar Radiation Atlas
LUT	Look-Up Table
MBE, RMBE	mean bias error, relative mean bias error
MFG	Meteosat first generation satellites
MSG	Meteosat second generation satellites
PV	photovoltaic
PVGIS	PhotoVoltaic Geographical Information System
PVGIS-3	previous version of PVGIS
PVGIS-CMSAF	new version of PVGIS with CM-SAF data
SRTM-3	Shuttle Radar Topography Mission digital terrain model

STD standard deviation

Symbols

CAL	effective Cloud ALbedo, also called cloud index (–)
D	diffuse horizontal irradiation (kWh/m ²)
G	global horizontal irradiance (W/m ²)
G_{clr}	global horizontal clear-sky irradiance (W/m ²)
H	global horizontal irradiation (kWh/m ²)
k	clear-sky index (–)
R	ratio of diffuse to global irradiation (–)

Subscripts

d	for given day
h	for given hour
m	for given month
y	for given year
$year$	long-term yearly average

(2003). Central applications discussed in the above cited paper are solar assessment and monitoring of photovoltaic systems (PV-SAT, pvsat.de) as well as Satel-Light (www.satel-light.com), an European database for solar radiation and daylight.

Heliosat-2 (Rigollier et al., 2004), a derivation of Heliosat has also been used within the SoDa service¹ (Wald et al., 2002) for the calculation of the solar surface irradiance. Satellite based solar irradiance derived with Heliosat (Heliosat-2) is also the basis for the SOLEMI service (Solar Energy Mining, www.solemi.de).

Solar surface irradiance derived from geostationary satellite measurements is usually more accurate than that interpolated from the ground-based measurements which are more than 30 km apart (Zelenka et al., 1999). This emphasizes the importance of satellite based irradiance, especially in sparsely populated regions.

Beside solar energy applications, satellite based solar irradiance is also used within the scope of climate monitoring and analysis. A widely used method within the climate community is that of Pinker and Laszlo (1992), which is based on relating the broadband transmission at the surface (T) to the broadband reflectance at the top of the atmosphere (R). The relationship between R and T is calculated with a radiative transfer model which accounts for the absorption by ozone and water vapor, multiple scattering by molecules, multiple scattering by aerosols and clouds, and multiple reflections between the surface and the atmosphere. Respective relations between R and T are saved in Look-up-Tables, which are subsequently used to estimate the solar irradiance for each satellite pixel and time step. The Pinker–Laszlo algorithm is used in the GEWEX

(Global Energy and Water Cycle Experiment) Surface Radiation Budget Project to generate the short-wave radiative fluxes (<http://gewex-srb.larc.nasa.gov/>).

Heliosat-like methods are usually applied to geostationary meteorological satellites. However, “Pinker–Laszlo” like methods can be also applied to polar-orbiting satellites which provides the possibility to extend the coverage of satellite based solar irradiance maps to regions at high latitudes (Wang et al., 2011; Hollmann et al., 2006; Wang and Pinker, 2009).

Solar radiation data are widely used for estimating the performance of solar energy systems. A number of solar radiation databases are available, some for free, while others are commercial products. See Šúri et al. (2008) for an intercomparison of some products, though others have appeared in the meantime, such as for instance SolarGIS (www.solargis.info). A number of performance estimation tools for photovoltaic (PV) systems can be found, some of which are stand-alone programs such as Meteonorm (www.meteonorm.com), PVsyst (www.pvsyst.com) and others, while some are web-based, such as SolarGIS or PVGIS. In the last few years PVGIS has emerged as a popular free web-based tool for quick estimates of PV performance in Europe and Africa.

In this paper we report on the use of a new satellite-based solar radiation data set in the PVGIS web-based system for estimating solar radiation and performance of PV systems. The main topic here is the validation of the new data set against ground station measurements, and the comparison between the old and new solar radiation data sets. The algorithms in PVGIS that use the solar radiation data, such as the PV performance models and the models for inclined-plane irradiation, are not affected by the change in source for the solar radiation data, and will not be discussed here.

¹ Integration and exploitation of networked Solar radiation Databases for environment monitoring project.

The organization of the paper is as follows: Section 2 will give an overview of the methods used for the calculation of solar irradiance from satellite data, including a brief overview of main validation results and references to previous work on the validation of the solar radiation estimates. Section 3 describes the PVGIS web application including the existing solar radiation database used in PVGIS. Section 4 gives a brief description of the methods used to incorporate the CM-SAF data into PVGIS. Section 5 reports on the validation of the new database for solar energy applications and describes the differences in the results from the two databases. Finally, Section 6 will present the conclusions and describe possible future work.

2. The CM-SAF solar radiation database

The CM-SAF solar surface irradiance reported in this paper is part of a suite of products derived within the Satellite Application Facility on Climate Monitoring (CM-SAF).² It contributes to the operational long term monitoring of the climate system by providing Essential Climate Variables related to the energy and water cycle of the atmosphere (e.g. surface and top of the atmosphere (TOA) radiation budget components) (Woick et al., 2002).

The CM-SAF solar radiation data base consists of solar irradiance data derived from the SEVIRI and GERB instruments on board the Meteosat Second Generation satellites (MSG) and from the MVIRI instrument on-board the Meteosat First Generation satellites (MFG). The algorithms applied to retrieve the solar surface irradiance differ between MSG and MFG.

For MSG the retrieval is based on the method of Pinker and Laszlo (1992). The method originally relates the top of atmosphere albedo to the surface irradiance for clear sky and cloudy cases. In the CM-SAF method this is only done for cloudy skies. For clear sky situation a clear sky model described in Mueller et al. (2009) is used, without explicit need of satellite data as input. The respective clear sky model, *gnuMAGIC*, has been reprogrammed in C and is available under GNU public license at (<http://sourceforge.net/projects/gnu-magic>).

The cloudy sky part of the operational CM-SAF hybrid-eigenvector LUT algorithm is not applicable to Meteosat First Generation (MFG). Instead, the well established Heliosat method (Cano et al., 1986) is used to consider the effect of clouds on the solar irradiance for the Meteosat First Generation satellites.

A modified version of Heliosat discussed in Posselt et al. (2011) is used to retrieve the effective cloud albedo (*CAL*, also called cloud index) from the observed reflections in the visible broadband channel.³ The effective cloud albedo is pre-dominantly linearly related to the solar irradiance by the following equation.

$$G = (1 - CAL)G_{clr} \quad (1)$$

This relation is a consequence of the law of energy conservation as e.g. discussed in Hay (2003). This relation is usually expressed with the clear sky index, which is the ratio between the actual (full-sky) surface solar irradiance (*G*) also denoted as global irradiance and the clear-sky surface solar irradiance (*G_{clr}*), namely,

$$k = \frac{G}{G_{clr}} \quad (2)$$

Hence, once the effective cloud albedo has been retrieved the solar irradiance can be derived by the use of the aforementioned clear sky model described in detail in Mueller et al. (2009).

The Heliosat method only uses the observed reflection for the treatment of the cloud effect on solar surface irradiance. Additional external information such as the surface albedo and cloud mask are not needed.

The operational CM-SAF products as well as the data sets have been validated against all available BSRN measurement sites. The accuracy is characterized by a mean absolute deviation of 8 W/m² (or better) and an accuracy better than 10 W/m² for 90% of all individual months, independent of the applied method and satellite generation. Relevant reviewed validation reports are public available at the CM-SAF web page (www.cmsaf.eu).

In addition to the solar surface irradiance (global irradiance) CM-SAF provides also direct irradiance for MSG and MFG. Here an adaptation of the approach of Skartveit et al. (1998) is used, which relates the clear sky index to the direct irradiance, please see Mueller et al. (2009) for further details.

3. The PVGIS solar radiation database and web-based PV estimation tool

3.1. Methodology of the existing PVGIS database, PVGIS-3

The construction and validation of the PVGIS solar radiation database has been described in a number of publications (Šúri et al., 2005, 2007), so a brief overview will suffice here. The PVGIS web applications can be found at <http://re.jrc.ec.europa.eu/pvgis/apps4/pvest.php>.

3.1.1. PVGIS-3 European database

The basis for the European radiation database within PVGIS is a collection of ground-based measurements made throughout Europe in the period 1981–1990 and collected and quality-checked as part of the European Solar Radiation Atlas (Scharmer and Greif, 2000). The dataset comprises a total of 560 stations in Europe and provides long-term averages of monthly global and diffuse horizontal irradiation, though in many cases the diffuse irradiation is estimated rather than measured.

From these spatial point data a continuous data set is produced by a 3D spatial interpolation technique (Mitasova

² www.cmsaf.eu

³ The effective cloud albedo (*CAL*) is also called cloud index in several publications, but herein we use the former term as this term defines the observed physical quantity.

and Mitas, 1993; Šúri and Hofierka, 2004). A cross-validation technique was employed (Šúri et al., 2007), in which each of the stations in turn was removed from the dataset and the interpolation repeated with the rest of the stations. The difference between this interpolation and that based on the full dataset gives an indication of the quality of the interpolation technique. The overall result showed that the cross-validation error was 4.5% for the entire dataset (Šúri et al., 2007). However, given the fact that the data set is spatially very heterogeneous, the actual error may be higher in areas with few stations or in areas with strong spatial variation in climate. Of course, the cross-validation also does not take into account any errors in the measurements themselves or in the postprocessing steps made within the ESRA project, such as modelling of diffuse irradiation.

3.1.2. PVGIS-3 African database

The PVGIS-3 solar radiation database is based on estimates of surface solar irradiance from satellite images. The calculations were performed within the HelioClim-1 Project (Blanc et al., 2011) using a version of the Heliosat-2 method (Rigollier et al., 2004) applied to satellite images from the geostationary Meteosat-3 to Meteosat-7 satellites. The full resolution of the satellite images was not used for the calculation and the resulting maps of solar radiation have a resolution of 15" over the entire field of view of the satellites. The database contains daily totals of global horizontal irradiation for the period 1985–2005 (Huld et al., 2005).

The original data with 15' resolution were downsampled to a 2 km grid by sampling and interpolating using the same 3D spatial interpolation technique used for the European database in PVGIS-3 (Mitasova and Mitas, 1993; Šúri and Hofierka, 2004).

It should be noted that the HelioClim-1 data used in PVGIS-3 were taken from one version of the database, and that the version described in Blanc et al. (2011) may not be completely identical to the one used in PVGIS-3.

3.2. Present capabilities of the PVGIS web application

The online interface to the PVGIS database lets the user estimate the long-term energy performance of different types of PV systems.

3.2.1. Grid-connected systems

The performance calculation for flat-plate grid-connected PV systems has the following features:

- For fixed systems, calculation of output for a given inclination and orientation (azimuth), or a calculation of the optimum inclination (and optionally orientation) (Šúri et al., 2005).
- For single-axis tracking systems, calculation of output for given inclination angles or optimization of inclination (Huld et al., 2010). Calculation of output for two-axis tracking systems.

- Calculation of the effects of temperature and irradiance on the performance of PV modules. This requires data also on ambient (air) temperature (Huld et al., 2008).
- Calculation of the effects of shadowing by terrain features. The underlying database of horizon height is calculated from the SRTM-3 digital elevation model (Farr et al., 2007). The horizon is pre-calculated for 48 directions using the original SRTM-3 spatial resolution of 3".

3.2.2. Stand-alone PV systems

The online calculation of stand-alone (off-grid) PV systems requires detailed time series of solar radiation data and has therefore until now only been available for Africa. The calculation of system performance uses the daily irradiation values for a number of years to calculate the overall performance for a given PV array (or module) and battery size and for a given desired energy consumption. The output consists of statistical information on the average energy produced, as well as the probability of energy supply failure due to empty battery and charging interruptions due to a fully charged battery.

Since the simulation is done over a number of years some of the features of the grid-connected estimates must be left out to save calculation time. Thus, it is not possible to calculate the optimum angle in this simulation.

4. Construction of the new PVGIS database from CM-SAF data

4.1. Spatial and temporal extent of data

The main source of data for the CM-SAF radiation dataset is the images of the Meteosat series of geostationary satellites. The overall extent of the images is approximately 70°N to 70°S and 70°W to 70°E. At the edges the uncertainty of the results is higher. For the first version of the CM-SAF-based version of PVGIS a more restricted spatial region was chosen, extending from 35°S to 58°N and from 20°W to 55°E. The time periods of the data used are:

- MFG-based data: 1998–2005, hourly values.
- MSG-based data: June 2006–December 2011 (starting October 2006 for the horizontal direct irradiance). Also here only one image per hour is used.

The global and direct irradiance have been calculated for each pixel in the satellite images. These data have subsequently been projected onto a latitude-longitude grid with spatial resolution of 1'30" for the MSG-based products and 1'48" for the MFG-based data.

For Europe the spatial resolution of PVGIS-CMSAF is somewhat lower (at 1'30") than the 1 km spatial resolution of PVGIS-3. However, since the high resolution of PVGIS-3 was obtained from interpolation between stations that are much more than 1 km apart, the PVGIS-CMSAF resolution can be regarded as being higher than that of

PVGIS-3, especially in regions where the solar radiation varies in a way that is not adequately captured by representative ground stations in the PVGIS-3 data set. For Africa, the resolutions of the old and new data set is similar, but again, the resolution of PVGIS-3 was artificially enhanced from the original 15' resolution of the HelioClim-1 data set.

4.2. Solar energy data requirements

The density of well maintained ground based measurements is quite scarce in many regions of the world, especially in arid and semi-arid regions with low population, but high solar energy potential. Solar surface irradiance data based on satellite observations is nowadays widely available. However, the validation results provided in Posselt et al. (2011) shows that established climate data sets like ISCCP (Rossow and Duenas, 2004) (International Satellite Cloud Climatology Project) and GEWEX (Gupta et al., 2006) (Global Energy and Water Cycle Experiment) are most likely not very usable for solar energy due to their coarse resolution and the large errors on local scale. This shows that the development of a retrieval scheme able to generate data in appropriate accuracy is a challenge. This compiles the development of the algorithm as well as the appropriate evaluation and selection of the needed input information of the atmosphere. Hence, scientific validation and evaluation is essential in order to proof the reliability of satellite derived data for solar energy applications. Here, different demands on performance of the data depends also on the solar energy system.

For long-term performance estimates of solar energy systems, the necessary solar radiation data can be summarized according to system type:

4.2.1. Grid-connected PV

For grid-connected PV systems, it can be assumed that the power produced can be immediately transferred to the electrical grid. The time of production is therefore not important for the estimate of overall output, and the estimate can be made with good accuracy using long-term monthly averages of global and diffuse irradiation. Some refinement of the PV performance estimates can be made if the variability of the solar radiation is known at least in a statistical sense (Huld et al., 2010). In the case of concentrating PV systems the necessary data consist of the long-term averages of direct normal irradiance (DNI).

4.2.2. Off-grid PV systems and domestic hot water systems

Both these types of systems store the energy locally for later local use. It is therefore possible that the energy storage may become full and further energy capture is not possible, or that the energy storage is emptied completely and the supply of energy is interrupted. To accurately model this it is necessary to know how the irradiance varies in time, at least on a daily basis, and preferably using hourly data. In addition, the efficiency of solar thermal hot water systems depends strongly on irradiance which makes it

necessary to know the probability of low-irradiance conditions.

The present capabilities of PVGIS are limited to making estimates of the performance of grid-connected flat-plate PV systems (whether fixed or sun-tracking), and (for Africa) estimates of off-grid PV systems. The availability of the solar radiation data from CM-SAF, as detailed in Section 2, will enable the following improvements and new features:

- Improvements in the estimates of grid-connected PV performance by a generally improved database for global and diffuse irradiation.
- Extension of the off-grid PV estimation capabilities to Europe.
- Extension of the PV estimation capabilities to include concentrating PV.

4.3. Calculation of long-term monthly averages

The calculation of the long-term monthly averages is complicated by the fact that not all hours are present in the data set. The MFG data set is nearly complete, with 97.4% of the hourly values present, while the MSG data set is somewhat less complete, at 93.3% of the hours present. When several days of a month are missing, the calculation of averages for that month becomes more problematic since the amount of solar radiation may change systematically during the month. To minimize this problem we perform the averaging in the following way: Consider the global irradiance map G_{hdm} for the hour h , day d , month m and year y . First the average over the years is calculated:

$$G_{hdm} = \frac{1}{N} \sum_{y=1}^N G_{hdm} \quad (3)$$

where N is the number of years for which the hour h , d , m is present. In this calculation of the average, there may be missing data for particular years, and the average is then performed over the years present. All hours during the year have at least one member present and nearly all have two or more members.

Next, the hourly values for each month are integrated in time to obtain the total long-term yearly average irradiation for that month, H_m . If the irradiation is expressed in Wh/m² (or kWh/m²), the integration becomes a simple sum:

$$H_m = \sum_{h=1}^{24} \sum_{d=1}^{D_m} G_{hdm} \quad (4)$$

where D_m is the number of days in month m . The monthly average irradiance is then:

$$G_m = \frac{H_m}{24D_m} \quad (5)$$

and the long-term average of the yearly total global horizontal irradiation is given as:

$$H_{\text{year}} = \sum_{m=1}^{12} H_m \quad (6)$$

This process has been performed independently for the global and direct components, and independently for the MFG and MSG data sets.

The MSG data set covers 5 years at present and the MFG data set 8 years. Due to the considerable year-to-year variation in solar irradiation a period of 5 years is too short to calculate a long-term average. It is therefore necessary to combine the two data sets. As will be shown in the following section, the MSG data set gives somewhat better results than the MFG data set, which is not surprising given the greater amount of information available from the MSG satellites, especially at infrared wavelengths. Thus, the MBE is generally lower for the MSG data set and more advanced algorithms are used to estimate irradiance in mountainous areas (Dürr and Zelenka, 2009). For these reasons it was decided that a long-term average data set would be constructed by a simple average of the monthly averages G_m for each of the two data sets.

5. Comparison of CM-SAF-based PVGIS with PVGIS-3

5.1. Validation and intercomparison of the solar radiation databases

5.1.1. Validation and intercomparison for Europe

The original PVGIS-3 for Europe and the new CM-SAF-based database do not cover the same time periods, indeed, they do not overlap at all. Therefore it is strictly speaking not possible to validate the two against each other. Differences between the two databases can of course

be due to errors in one or both databases, but it may also be an indication of a change in the climate over the last 30 years. In particular, the aerosol load in the atmosphere in Europe may have changed. Still, a difference between the two databases due to a change in the climate may not mean that the PVGIS-3 is “wrong”, but *would* mean that it is no longer up-to-date for its stated purpose of predicting solar energy output for the future.

One possible comparison is to look at the sites for which high-quality ground station data are available and compare the validation results from CM-SAF with the difference between CM-SAF and PVGIS-3. Table 1 shows the relative MBE of the for CM-SAF (MFG and MSG) for a number of locations, and for the same locations the difference between the new CM-SAF-based PVGIS with the PVGIS-3 database. A couple of stations do not have data for the period covered by MFG, and the station in Sde Boquer is outside the PVGIS-3 area.

It should be emphasized that in Table 1 the validation is generally done with 1–2 years of data, which do not necessarily cover the same time periods for the different stations. In contrast, the comparison of PVGIS-3 and PVGIS-CMSAF uses the full data set of both databases.

The Mean bias error varies strongly between locations, with absolute values ranging from almost nothing to more than 15%. To get an indication of the uncertainty of the annual irradiation estimates, the standard deviation of the RMBE values was calculated. For the MSG data set, STD (RMBE) = 5.3% while for the MFG data set is slightly higher at STD (RMBE) = 5.5%. The new database is constructed by simple averaging of these two data sets, so the combined RMBE for each station is the average of the MBE's for each data set. The standard deviation of

Table 1

Comparison of CM-SAF MBE results with the change from the PVGIS-3 to the new CM-SAF PVGIS over Europe. The MBE values are relative values for the global horizontal irradiation.

Station	Latitude	Longitude	RMBE MSG (%)	RMBE MFG (%)	Diff. CM-SAF vs. PVGIS-3 (%)
Lindenberg (DE)	52.22N	14.12E	−3.4	−3.0	+7.0
Cabauw (NL)	51.97N	4.93E	+0.4	+1.5	+11.6
Carpentras (FR)	44.05N	5.03E	+2.1	+5.1	+9.1
Payerne (CH)	46.81N	6.94E	−3.0	+3.7	+13.2
Camborne (UK)	50.22N	5.32W	+3.0	+6.2	+8.4
Toravere (EE)	58.27N	26.47E	+5.1	+4.5	0.0
Sde Boquer (IL)	30.87N	34.77E	−3.3	+5.0	–
Almeria (ES)	37.50N	2.2W	−0.9	+2.7	+9.3
Geneve (CH)	46.12N	6.01E	+2.6	+6.2	+8.3
Nantes (FR)	47.25N	1.55W	+3.8	+3.8	+6.7
Vaulx-en-Velin (FR)	45.78N	4.93E	+3.9	+9.0	+6.6
Kishinev (MO)	47.0N	28.82E	+0.4	+3.0	+3.2
Liepaja (LV)	56.48N	21.02E	+2.5	+2.0	+6.2
Sonnblick (AT)	47.05N	12.95E	−14.0	−16.7	−13.8
Thessaloniki	40.63N	22.97E	+5.9	+3.5	+17.6
Wien H. Warte (AT)	48.25N	16.35E	−1.5	+3.5	+3.8
Ispira (IT)	45.81N	8.64E	+8.4	+8.6	+15.0
Milano (IT)	45.48N	9.26E	−0.5		+13.0
Roma (IT)	41.86N	12.62E	+4.1	−0.9	+11.1
Sarreguren (ES)	42.82N	1.60W	+1.66		+5.5
A Coruña (ES)	43.37N	8.42W	+11.0	+2.9	+11.2
LLeida (ES)	41.62N	0.595E	+2.4	−0.1	+16.7
Madrid (ES)	40.45N	3.72W	−0.3	+6.6	+7.5

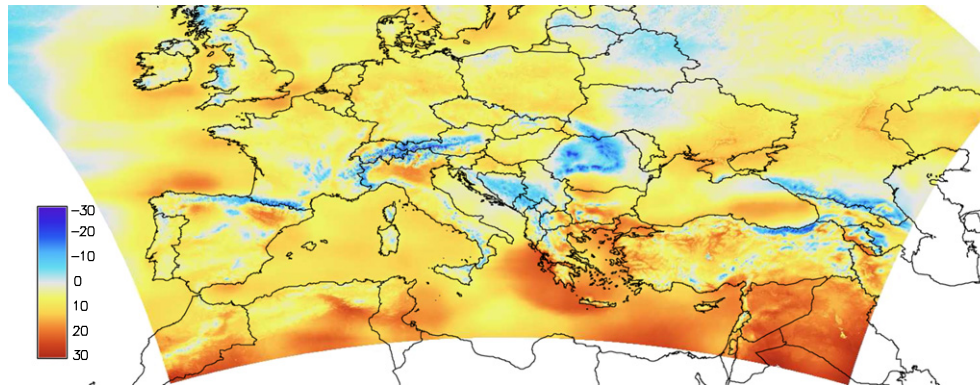


Fig. 1. Relative difference in the yearly average global horizontal irradiation between CM-SAF and the European part of PVGIS-3. The difference is expressed in percent over the area covered by both databases.

the station MBE's in the combined data set is STD (RMBE) = 4.9%.

One station with very high RMBE is Sonnblick, which is on a mountaintop at 3105 m altitude. This location is not very representative of the locations where PV systems are normally installed. Excluding this station, the overall STD (RMBE) = 3.8%. The poor result in this location does however highlight the need for improvements in mountainous areas.

From Table 1 it is clear that in general the RMBE of both CM-SAF databases are significantly smaller than the difference between PVGIS-3 and PVGIS-CMSAF. Notable exceptions are Sonnblick in the high mountains where the RMBE of PVGIS-CMSAF is larger than the difference between the databases, Ispra in Northern Italy, Toravere, Estonia, and A Coruña, Spain. In Ispra, the difference in estimates is 15.0% while the RMBE of PVGIS-CMSAF is about 8.5%, indicating that the new database overestimates the irradiation by about as much as the old database underestimates. In Toravere, both databases give the same value, but PVGIS-CMSAF has a positive RMBE of 4.5%, which seems to indicate that both databases may give too high values here. The station in A Coruña is only a few hundred meters from the coast, which means that the satellite pixel “sees” a mixture of land and sea. This may lead to errors in the solar radiation calculation, which indicates that the uncertainties in PVGIS-CMSAF may be larger very close (less than about 2 km) from the coast.

It is also evident that in all the validation sites PVGIS-3 has lower yearly global horizontal irradiation, except the mountain station at Sonnblick.

Fig. 1 shows a map of the difference between the old PVGIS-3 and the new CM-SAF based database, PVGIS-CMSAF. The difference is calculated as the relative difference in H_{year} (see Eq. (6)). Over large areas in Central Europe the difference is fairly uniform at about +5–8%. However, there are areas where the differences are larger. The new dataset shows a large increase on the old PVGIS-3 in the Po Plain in Northern Italy, in most of Catalonia, Spain and in Bulgaria. On the other hand, the new dataset gives

lower values in most mountainous areas, in particular the Alps and the Carpathians, as well as in parts of Romania. In Great Britain the pattern of differences show that in the east PVGIS-CMSAF gives higher values while in the west PVGIS-CMSAF shows lower values than PVGIS-3.

5.1.2. Discussion of differences, Europe

As seen in Fig. 1, the general tendency is an increase in global horizontal irradiation from PVGIS-3 to the new dataset. Validation exercises have shown that the CM-SAF datasets have a rather low overall MBE, at -1 W/m^2 for the MSG data set and 3.9 W/m^2 for the MFG data set (Mueller et al., 2009). These values correspond to about 3% in Northern Europe and less than 2% in Southern Europe. As is seen in Table 1 the differences from PVGIS-3 to PVGIS-CMSAF are larger than that at the large majority of validation points. This indicates that PVGIS-CMSAF generally improves accuracy relative to PVGIS-3.

The fact that some large areas show a rather uniform difference (for example most of lowland Central Europe) points to a systematic difference between the two data sets in these areas. The validation results for these areas show that the CM-SAF data sets are generally reasonably accurate. Therefore it must be concluded that the PVGIS-3 values are too low in these regions. Interpolation cannot produce large biases over extended regions, so the reason must be due to the measurement station data. It is of course possible that some stations will report too low values over long time intervals. There are many effects that would produce a negative bias in measurements, such as snow, dirt or shadows. However, it seems unlikely that many stations would have a similar negative bias in the measurements.

Another possibility is that the difference between the two data sets reflects a change in the global horizontal irradiation over the time period between the time periods of the two data sets. Several authors have discussed the “global dimming” taking place in the second half of the 20th century, followed by “global brightening” since the 1980's (Wild, 2009; Wild et al., 2005). In Wild (2009) some trend values for Europe are given. For Europe as a whole, global

horizontal irradiation increased by approximately 2.5% per decade over the period 1985–2005. In Germany and the Low Countries the trend was somewhat larger, at 4%/decade. This change over time in irradiation can account for most of the general difference between PVGIS-3 and the new PVGIS-CMSAF. However, as is clearly seen in Fig. 1, it can not be used to explain local differences.

If the changes over time in irradiation are a real physical effect, it is not really correct to speak of errors in PVGIS-3. However, the main use of the PVGIS database is to make estimates of solar energy system performance for the present and the future. Hence it is more useful to know the present level of irradiation rather than a value that was valid 20–30 years ago.

Apart from the overall difference between the two data sets there are also particular patterns in some areas. The difference between PVGIS-CMSAF and PVGIS-3 is particularly large in The Po Plain in Northern Italy and in Catalonia (Spain). Such local differences may be due to errors in one or a few measurement stations, or it may reflect a problem with the new data set. In the case of the Po Plain we have used data from Milano (45.48N, 9.26E) to validate the CM-SAF MSG data set for the year 2009. Here it was found that the MBE is only -0.5% , indicating that the new data set performs well in that area and that it is almost certainly the old data set that is wrong. Only 70 km away, in Ispra in the Italian lake region, the situation is very different. While the difference between the new and the old database is $+15\%$, the MBE of PVGIS-CMSAF is also high, at about 8.5% for both the MFG and the MSG part. In this case the algorithm calculating solar radiation may have problems because of the complex terrain features, with a single pixel in the satellite image covering built-up areas, forest, and lakes. In this instance, the likely error of PVGIS-CMSAF is at least as high as that of PVGIS-3.

Among other features of interest can be noted Romania and Bulgaria where the two data sets have significant differences. In this area the PVGIS-3 is based on a very sparse set of measurement stations which make it likely that important climatic features are not properly represented. In Great Britain there is a clear difference from east to west, with PVGIS-CMSAF giving higher values in the eastern half of the island but lower values in PVGIS-3. Here there is a well-known difference between the more cloudy western half and the drier eastern half. It is possible that the interpolation method of PVGIS-3 has not captured this gradient properly due to a lack of representative measurement stations. However, this area is also relatively far north, where the uncertainty of the CM-SAF is known to be higher.

Finally, mountain areas show large differences, with PVGIS-3 generally showing higher values, except at very high elevations above 3000 m. In this case both databases have known weaknesses. In the case of PVGIS-3 the interpolation may have problems due to the low number of radiation measurements at high elevation. For the satel-

lite-based method of CM-SAF (and hence PVGIS-CMSAF), it is well-known that radiation estimates from satellite images have difficulties in distinguishing snow/ice from clouds (Dürr and Zelenka, 2009). In this case it is more difficult to state with confidence which of the two databases gives the more accurate result. For the single measurement location in Table 1 at high altitude, PVGIS-CMSAF shows very high error at about $+15\%$, so in this case it is likely that the old database would be more accurate. However, it is not possible to draw more general conclusions from a single measurement station.

5.1.3. Comparison for Africa

For Africa, the existing PVGIS dataset and the CM-SAF dataset partially overlap in time. They are also partially derived from the same satellite image data. Hence, it should be possible to perform a validation of the two data sets for the same time period. Unfortunately, for Africa the number of sites with good ground station measurements is very limited.

A few stations with high quality radiation data are available from the BSRN network in the Africa and Middle East area. For these four stations we have validated the long-term global horizontal irradiation. The relevant parameter here is the Mean Bias Error. Table 2 shows the validation results for these four stations. The results show that for the two stations in Africa (Tamanrasset and De Aar) both databases give reasonable results. However, for the other two stations, located in the Middle East, the new database has much lower bias error than the old PVGIS-3 database.

Fig. 2 shows the yearly average global horizontal irradiation for PVGIS-3. Fig. 3 shows the difference between the PVGIS-CMSAF and PVGIS-3. Although the spatial pattern of the two are similar in most areas (for instance the lower irradiation in the tropical belt), there are also significant differences. The most prominent difference is the area in southwestern Sahara (running across Mali, Mauritania and parts of Burkina Faso), where the irradiation of the new PVGIS-CMSAF database is considerably higher than PVGIS-3. Another region of large differences is the coastal areas of Egypt and the Nile Delta, as well as parts of the Arabian Peninsula.

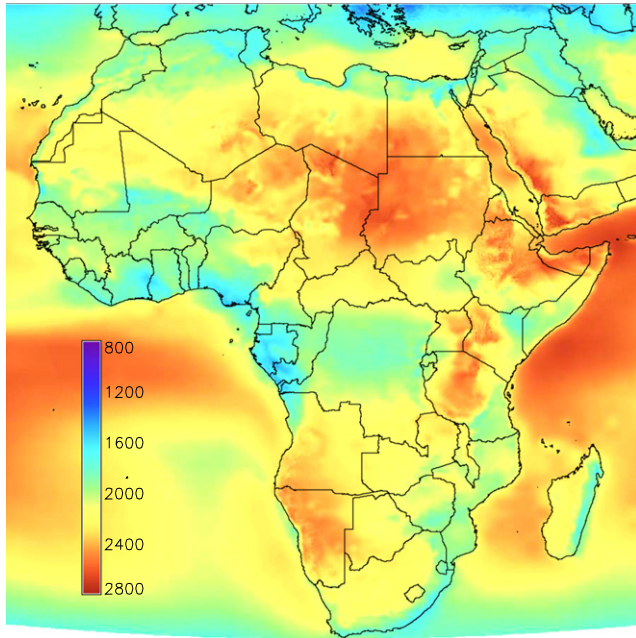
For the Arabian Peninsula, the difference is also seen in the difference in bias error at the Solar Village measurement station in Saudi Arabia. For the other areas it was not possible to find any suitable data for validation. However, there are reasons to believe that the problem lies with the older database. The area in Southwestern Sahara is characterized by very high ground albedo and it is possible that the Heliosat-2 algorithm used for the calculation of the solar radiation has problems distinguishing between clouds and the very bright ground.

In some mountain areas in East Africa the new PVGIS-CMSAF values are much lower than PVGIS-3. This is likely to be an artifact of the downscaling procedure and

Table 2

Relative mean bias errors for PVGIS-3 and PVGIS-CMSAF for four stations in Africa and the Middle East. Errors are given in percentage points.

Location	Latitude	Longitude	RMBE CM-SAF (%)	RMBE (%) PVGIS-3 (%)
Tamanrasset (DZ)	22°46'48"N	5°30'36"E	−6.0	−0.4
De Aar (ZA)	30°40'S	23°59'35"E	+2.2	−1.8
Sde Boqer (IL)	30°54'18"N	34°46'55"E	+4.0	−13.9
Solar Village (SA)	24°54'36"N	46°24'36"E	+3.2	−14.8

Fig. 2. Map of average annual global horizontal irradiation using PVGIS-3, given in kWh/m².

does not represent a difference between the original HelioClim-1 data and the CMSAF data.

Further validation results for HelioClim-1 can be found in a number of publications, see for instance Abdel Wahab et al. (2009) and Wald et al. (2011). Note however, as mentioned in Section 3.1.2, the version of HelioClim-1 used in PVGIS-3 may not be completely identical to that used in these papers.

5.2. Differences in derived parameters

The most important component of the databases is the global horizontal irradiation. However, the performance of PV systems is not fully determined just by the global horizontal radiation. Since the PV modules are normally mounted at an angle (inclination) from horizontal and may be mounted on tracking systems, also the amount of diffuse irradiation becomes important.

The distribution of irradiation over the year also has an influence on the performance of PV systems, since it determines (together with the diffuse irradiation) the optimum inclination angle for fixed (non-tracking) PV systems.

Fig. 4 shows the ratio R of horizontal diffuse irradiation D_{year} to global irradiation: $H_{\text{year}} R \equiv D_{\text{year}}/H_{\text{year}}$, together

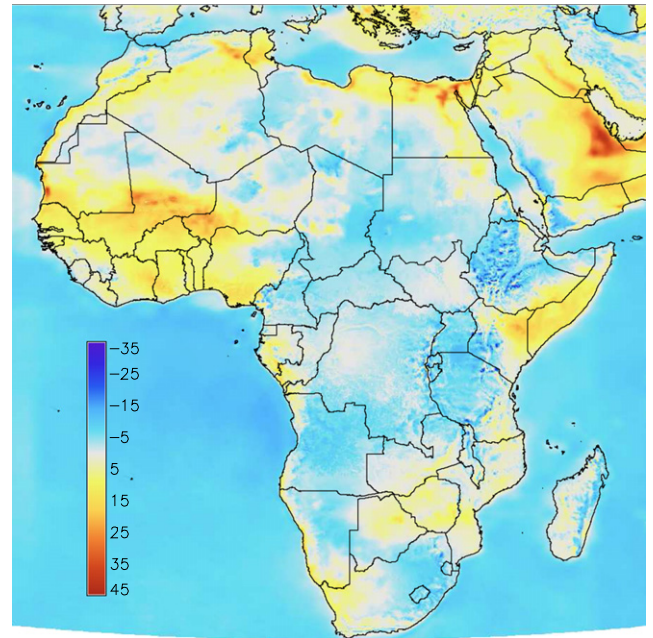


Fig. 3. Relative difference in annual global horizontal irradiation between PVGIS-CMSAF and PVGIS3. The difference is given in %.

with the absolute difference between the diffuse/global ratio for the two databases.

Comparing Fig. 1 with Fig. 4b, it is easy to see that the two are almost mirror images of each other. In most of Europe the new PVGIS-CMSAF data show higher irradiation and corresponding lower D/G , while in mountainous areas the lower irradiation is accompanied by a higher R . However, in some areas, such as western Great Britain, Bosnia and the Carpathians, there is a slight decrease in overall irradiation, but R is almost unchanged.

The general lowering of the diffuse fraction may have an influence on the calculation of the optimum angle for fixed-mounted PV systems. Generally, the optimum angle is a trade-off between capturing as much as possible of the direct sunlight, which would indicate a mounting inclination angle almost equal to the latitude, and capturing the diffuse irradiation, which would call for an almost horizontal mounting. In PVGIS-3 it was found that the optimum angle range between 30° and 35° from horizontal over much of Europe, due to the gradual increase of R with latitude (Šúri et al., 2005).

The calculation of optimum angle is strongly affected by shadows (Šúri et al., 2005), and a comparison between the old and new PVGIS versions should therefore be made

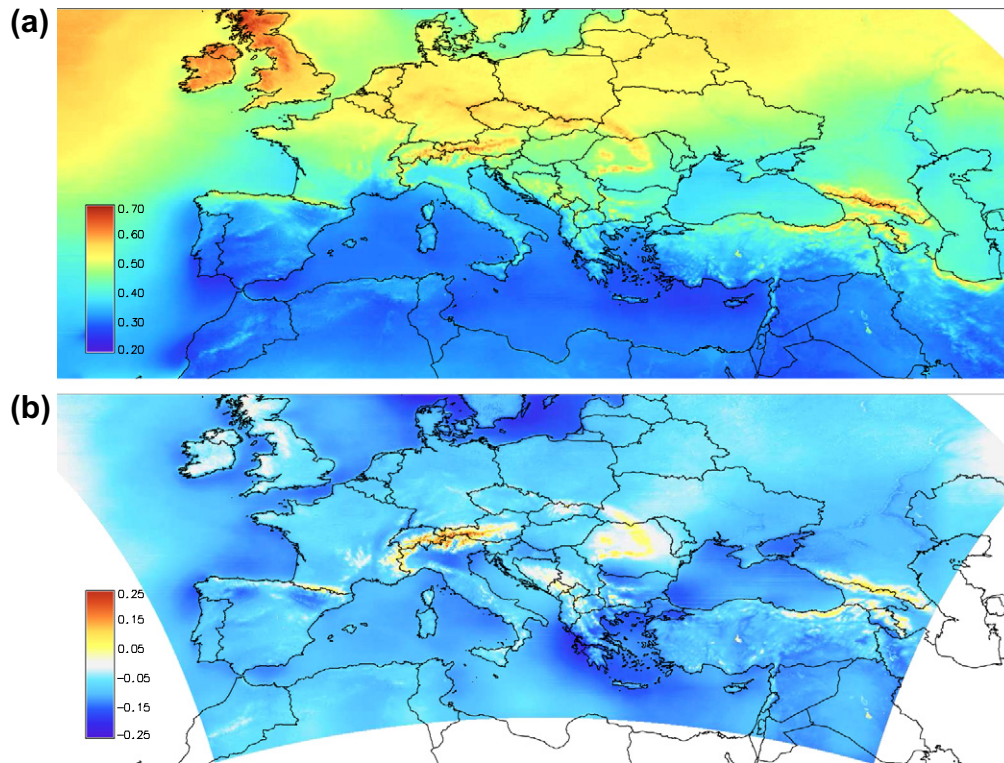


Fig. 4. (a) Ratio of yearly horizontal diffuse irradiation to global irradiation for PVGIS-CMSAF in Europe. (b) Absolute difference from PVGIS-3 to PVGIS-CMSAF.

with the same digital elevation model. For this reason it was decided to perform the calculation of the optimum angle in the map projection hitherto used in PVGIS: Lambert Azimuthal Equal Area with centerpoint 48°N 18°E and a resolution of 1 km. The solar radiation data from PVGIS-CMSAF were projected onto this grid for the optimum angle calculation.

Fig. 5 shows the optimum angle for fixed-mounted south-facing PV modules in Europe, calculated using PVGIS-CMSAF and the method described in Šuri et al. (2005), together with the difference in the optimum angle between PVGIS-3 and PVGIS-CMSAF.

In most locations the difference in optimum angle is not large, between -5° and $+5^\circ$. Close to the optimum angle, the yearly irradiation changes only very slightly as the angle changes, so a change of less than 5° will generally result in a change of annual irradiation of less than 1%. However, especially in the Alps, the optimum angle as calculated from PVGIS-CMSAF is significantly lower than the old values. The general trend is that areas with increased irradiation now have a steeper optimum while areas with decreased irradiation have lower optimum angles. This fits well with the observation that the diffuse to global ratio R is lower in PVGIS-CMSAF in the areas where the new data show higher irradiation. When the diffuse contribution is small, the optimum angle will be steeper to better capture the direct sunlight.

5.2.1. Diffuse irradiation over Africa

A comparison of R for Africa is shown in Fig. 6.

While in Europe increased irradiation is associated with a decreased R , in Africa the relation is less clear. In general, the new database shows higher diffuse fraction in most places. The exceptions are the areas that show the strongest increase in annual irradiation: Southwest Sahara, and Northern Libya and Egypt, where R is lower or unchanged. For Africa, the diffuse irradiation in PVGIS-3 was estimated from long-term irradiation values using a very simple model, which in some regions gave very low values for the diffuse irradiation, barely higher than that found from a calculation of clear-sky values. Therefore it is likely that the new database represents an improvement in the estimate of R , although the small number of measurement stations makes it difficult to assess this accurately. For fixed-mounted PV systems in Africa, the optimum angle is relatively low, so an error in R only has a modest effect on the accuracy of the production estimate. For tracking systems, which spend part of the time with the modules at very steep angles, errors in R will be more important.

6. Conclusions and future plans

Thanks to the free availability of solar radiation data from the Climate Monitoring SAF it has been possible to construct a new spatial database of solar irradiation values for inclusion into the PVGIS database and web application. The new data set has been shown to be of high quality and will increase the accuracy of the PV performance estimates calculated by PVGIS. A validation of the global horizontal irradiation using data from 20 stations has shown

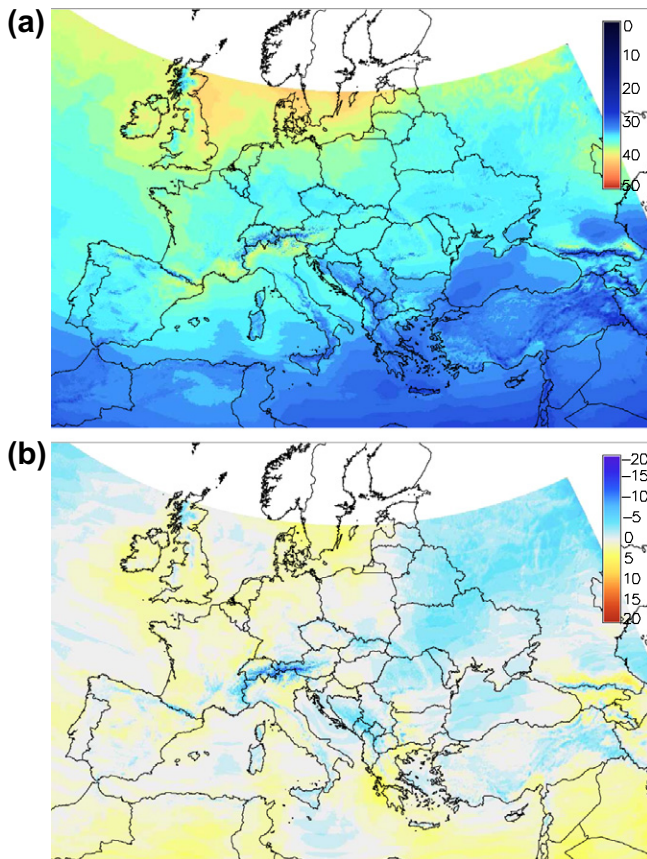


Fig. 5. (a) Optimum angle for fixed-mounted south-facing PV modules as calculated using the PVGIS-CMSAF database. (b) Difference in the optimum angle between PVGIS-CMSAF and that calculated using PVGIS-3.

that the overall Mean Bias Errors of PVGIS-CMSAF is low, at about +2.0%, while the standard deviation of individual station MBE values is 5.0%.

The PVGIS-CMSAF database is already available in the PVGIS web application. For backwards compatibility the old PVGIS-3 database is still available (users choose between the databases in the web interface). This will also enable users to make a comparison between the databases for the locations of interest.

The spatial coverage of the new database is somewhat more restricted than the existing PVGIS database. In particular, the region north of 58°N is not covered by PVGIS-CMSAF, mainly due to the high uncertainty of radiation estimates from geostationary satellite data at high latitudes. This problem will require more work, either on improvements to the algorithms for geostationary data or by using other data sources, such as polar-orbiting satellite data.

Mountain areas present show a high level of difference between the two databases and it is not entirely clear if the new database is significantly more accurate than the old one. In this case more detailed validation would be useful and will be performed subject to the availability of suitable ground station measurements. A special version of CM-SAF radiation data has been prepared for the Alpine region using the methods proposed by Dürr and Zelenka

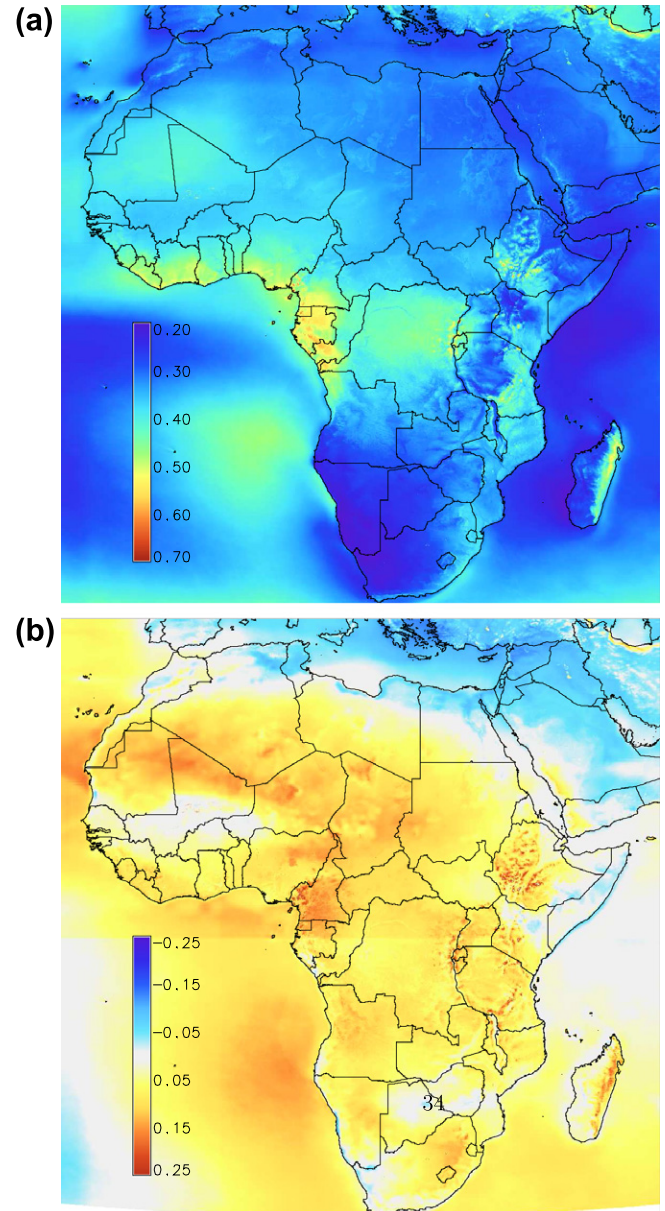


Fig. 6. (a) Ratio of yearly horizontal diffuse irradiation to global irradiation, R , for PVGIS-CMSAF in Africa. (b) Absolute difference from PVGIS-3 to PVGIS-CMSAF.

(2009). This data set may be incorporated into PVGIS in the near future.

CM-SAF provides hourly data of solar irradiance over both Europe and Africa. These data will make it possible to extend the calculation of off-grid systems to Europe. It will also make it possible to make a more detailed calculation of off-grid systems than the one presently possible with daily irradiation data, and to check the accuracy of the algorithm used in the off-grid web application of PVGIS.

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References

- Abdel Wahab, M., El Metwally, M., Hassan, R., Lefèvre, M., Oumbe, A., Wald, L., 2009. Assessing surface solar irradiance in Northern Africa desert climate and its long-term variations from Meteosat images. *Int J Remote Sens* 31 (01), 261–280. <http://dx.doi.org/10.1080/01431160902882645>.
- Blanc, P., Gschwind, B., Lefèvre, M., Wald, L., 2011. The HelioClim Project: surface solar irradiance data for climate applications. *Remote Sens.* 3, 343–361. <http://dx.doi.org/10.3390/rs3020343>.
- Cano, D., Monget, J., Albuissou, M., Guillard, H., Regas, N., Wald, L., 1986. A method for the determination of the global solar radiation from meteorological satellite data. *Sol Energ* 37, 31–39.
- Dagestad, K.-F., 2004. Mean bias deviation of the Heliosat algorithm for varying cloud properties and sun-ground-satellite geometry. *Theor. Appl. Climatol.* 79, 215–224. <http://dx.doi.org/10.1007/s00704-004-0072-5>.
- Dürr, B., Zelenka, A., 2009. Deriving surface global irradiance over the Alpine region from METEOSAT second generation data by supplementing the HELIOSAT method. *Int. J. Remote Sens.* 30 (22), 5821–5841.
- Farr, T.G. et al., 2007. The shuttle radar topography mission. *Rev. Geophys.* 45, RG2004. <http://dx.doi.org/10.1029/2005RG000183>.
- Gupta, S.K., Stackhouse Jr., P.W., Cox, S.J., Mikovitz, J.C., Zhang, T., 2006. 22-Year surface radiation budget data set. *GEWEX News* 16 (4), 1213.
- Hammer, A., Heinemann, D., Hoyer, C.R.K., Lorenz, E., Mueller, R., Beyer, H., 2003. Solar energy assessment using remote sensing technologies. *Remote Sens. Environ.* 86, 423–432.
- Hay, John E., 2003. Satellite based estimates of solar irradiance at the Earth's surface – I. Modelling approaches. *Renew Energ* 3 (4,5), 381–393.
- Hollmann, R., Müller, R.W., Gratzki, A., 2006. CM-SAF surface radiation budget: first results with AVHRR data. *Adv. Space Res.* 37, 2166–2171.
- Huld, T., Šúri, M., Dunlop, E., Albuissou, M., Wald, L., 2005. Integration of HelioClim-1 database into PV-GIS to estimate solar electricity potential in Africa. In: *Proceedings of the 20th European Photovoltaic Solar Energy Conference*, Barcelona, Spain, pp. 2989–2992.
- Huld, T., Šúri, M., Dunlop, E.D., 2008. Geographical variation of the conversion efficiency of crystalline silicon photovoltaic modules in Europe. *Progr. Photovolt. Res. Appl.* 16 (7), 595–607.
- Huld, T., Gottschalg, R., Beyer, H.-G., Topić, M., 2010. Mapping the performance of PV modules, effects of module type and data averaging. *Sol Energ* 84, 324–338.
- Huld, T., Šúri, M., Cebecauer, T., Dunlop, E.D., 2010. Analysis of one-axis tracking strategies for PV systems in Europe. *Progr. Photovolt. Res. Appl.* 18 (3), 183–194.
- Ineichen, P., Perez, R., 1999. Derivation of cloud index from geostationary satellites and application to the production of solar irradiance and daylight illuminance data. *Theor. Appl. Climatol.* 64, 119–130.
- Ineichen, P., Barroso, C.S., Geiger, B., Hollmann, R., Marsouin, A., Mueller, R., 2009. Satellite application facilities irradiance products: hourly time step comparison and validation over Europe. *Int. J. Remote Sens.* 30, 5549–5571.
- Mitasova, H., Mitas, L., 1993. Interpolation by regularized spline with tension: I. Theory and implementation. *Math. Geol.* 25, 641–655.
- Mueller, R., Matsoukas, C., Gratzki, A., Behr, H., Hollmann, R., 2009. The cm-saf operational scheme for the satellite based retrieval of solar surface irradiance – a lut based eigenvector hybrid approach. *Remote Sens. Environ.* 113 (5), 1012–1024.
- Perez, R., Seals, R., Zelenka, A., 1997. Comparing satellite remote sensing and ground network measurements for the production of site/time specific irradiance data. *Sol. Energ.* 60, 89–96.
- Perez, R., Aguiar, R., Collares-Pereira, M., Dumortier, D., Estrada-Cajigal, V., Gueymard, C., Ineichen, P., Littlefair, P., Lund, H., Michalsky, J., Olseth, J., Renne, D., Rymes, M., Skartveit, A., Vignola, F., Zelenka, A., 2001. Solar resource assessment: a review. In: *Solar Energy – The state of the Art*. No. ISBN: 1 902916239 in ISES Position Papers. James & James Science Publishers, London, pp. 497–562.
- Pinker, R., Laszlo, I., 1992. Modelling surface solar irradiance for satellite applications on a global scale. *J. Appl. Meteor.* 31, 166–170.
- Posselt, R., Müller, R., Stöckli, R., Trentmann, J., 2011. Spatial and temporal homogeneity of solar surface irradiance across satellite generations. *Remote Sens.* 3, 1029–1046. <http://dx.doi.org/10.3390/rs3051029>.
- Rigollier, C., Lefevre, M., Wald, L., 2004. The method Heliosat-2 for deriving shortwave solar radiation from satellite images. *Sol. Energ.* 77, 159–169.
- Rossov, W.B., Duenas, E., 2004. The International Satellite Cloud Climatology Project (ISCCP) web site: an online resource for research. *Bull. Amer. Meteorol. Soc.* 85, 167–172. <http://dx.doi.org/10.1175/BAMS-85-2-167>.
- Skartveit, A., Olseth, J., Tuft, M., 1998. An hourly diffuse fraction model with correction for variability and surface albedo. *Sol. Energ.* 63, 173–183.
- Scharmer, K., Greif, J. (Eds.), 2000. *The European Solar Radiation Atlas, vol. 2: Database and Exploitation Software*. Les Presses de l'Ecole de Mines, Paris.
- Šúri, M., Hofierka, J., 2004. A new GIS-based solar radiation model and its application to photovoltaic assessments. *Trans. GIS* 8, 175–190.
- Šúri, M., Huld, T., Dunlop, E.D., 2005. PV-GIS: a web-based solar radiation database for the calculation of PV potential in Europe. *J. Sustain. Energ.* 24, 55–67.
- Šúri, M., Huld, T., Dunlop, E.D., Ossenbrink, H., 2007. Potential of solar electricity generation in the European Union member states and candidate countries. *Sol. Energ.* 81, 1295–1305.
- Šúri, M., Remund, J., Cebecauer, T., Dumortier, D., Wald, L., Huld, T., Blanc P., 2008. First steps in the cross-comparison of solar resource spatial products in Europe. In: *Proceedings of EUROSUN 2008*, Lisbon, Portugal, 7–10 October 2008.
- Wald, L., Albuissou, M., Best, C., Delamare, C., Dumortier, D., Gaboardi, E., Hammer, A., Heinemann, D., Kift, R., Kunz, S., Lefevre, M., Leroy, S., Martinoli, M., Menard, L., Page, J., Prager, T., Ratto, C., Reise, C., Remund, J., Rimoczi-Paal, A., Van der Goot, E., Vanroy, F., Webb, A., 2002. Soda: a project for the integration and exploitation of networked solar radiation databases. In: *Pillmann, W., Tochtermann, K. (Eds.), Environmental Communication in the Information Society. Part 2. International Society for Environmental Protection*, Vienna, Austria, pp. 713–720.
- Wald, L., Blanc, Ph., Lefvre, M., Gschwind, B., The performances of the HelioClim databases in Mozambique. In: *ISES Solar World Congress 2011*, Kassel, Germany, 28 August–2 September 2011.
- Wang, H., Pinker, R.T., 2009. Shortwave radiative fluxes from MODIS: model development and implementation. *J. Geophys. Res.* 114, D20201. <http://dx.doi.org/10.1029/2008JD010442>.

- Wang, P., Stammes, P., Müller, R., 2011. Surface solar irradiance from SCIAMACHY measurements: algorithm and validation. *Atmos. Meas. Tech.* 4, 875–891.
- Wild, M., Gilgen, H., Roesch, A., Ohmura, A., Long, C.N., Dutton, E.G., Forgan, B., Kallis, A., Russak, V., Tsvetkov, A., 2005. From dimming to brightening: decadal changes in solar radiation at Earth's surface. *Science* 308, 847–850.
- Wild, M., 2009. Global dimming and brightening: a review. *J. Geophys. Res.*, 114. <http://dx.doi.org/10.1029/2008JD011470>.
- Woick, H., Dewitte, S., Feijt, A., Gratzki, A., Hechler, P., Hollmann, R., Karlsson, K.-G., Laine, V., Löwe, P., Nitsche, H., Werscheck, M., Wollenweber, G., 2002. The satellite application facility on climate monitoring. *Adv. Space Res.* 30 (11), 2405–2410.
- Zelenka, A., Perez, R., Seals, R., Reme, D., 1999. Effective accuracy of satellite-derived hourly irradiances. *Theor. Appl. Climatol.* 62, 199–207.